Document tabs
Tab 1
Technical Instructions for...
1. API Endpoint & WebS...
2. System Prompt & Per...
3. Scalability Across Us...

Technical Instructions for CompanySync / YICN Voice Assistant Engine

To: Alex / Lexi / Development Team
Subject: Required Setup & Configuration for Low-Latency Voice Assistant Pipeline

1. API Endpoint & WebSockets Architecture

- **Use Gemini Live API (WebSocket):** Switch connection models to use Google's native WebSocket streaming endpoint (e.g., gemini-3.1-flash-live-preview or gemini-live-2.5-flash-native-audio).
- **Native Audio-to-Audio:** Do **not** use a traditional 3-step chain (Speech-to-Text → Gemini Text → Third-Party TTS). Send raw audio directly over the WebSocket to process native audio input/output and achieve sub-second response times.
- **Enable Full-Duplex / Barge-In:** Turn on Voice Activity Detection (VAD) and interruption handling so that if the user cuts in mid-sentence, the AI cuts its audio stream instantly to listen.
- **Output Limit:** Set max_output_tokens to **100–150 per turn** to guarantee short, rapid-fire conversational bursts rather than long paragraphs.

2. System Prompt & Persona Rules

Incorporate the following strict rules into the base system prompt:

```
You are a live phone assistant for CompanySync / YICN.

VOICE & BEHAVIOR RULES:
1. BREVITY: Keep every response to 1 to 2 short sentences. Never give long speeches or bulleted lists.
2. NATURAL FLOW: Speak like a real person over the phone. Use natural, conversational language (e.g., "Gotcha," "Makes sense," "No problem").
3. ONE QUESTION AT A TIME: Always end your turn with a single, clear question, then STOP speaking and wait for the user to reply.
4. HANDLING INTERRUPTIONS: If the user cuts in or speaks over you, stop immediately and address what they just said.
5. INTENT: Be helpful, direct, and focused on guiding the conversation to the next step
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Tab 1
Technical Instructions for...
1. API Endpoint & WebS...
2. System Prompt & Per...
3. Scalability Across Us...

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without sounding rigid or scripted.
```

3. Scalability Across Use Cases

Once this core pipeline and latency threshold are established, maintain the same technical architecture for all three operational tiers by updating only the secondary task persona instructions:

- **Outbound Calls:** Lead follow-ups, appointment setting, and demo bookings.
- **Inbound Customer Service:** Customer updates, claims inquiries, and message routing.
- **In-House Team Operations:** Field logging, delivery updates, and hands-free task completion.

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1. API Endpoint & WebS...
2. System Prompt & Per...
3. Scalability Across Us...